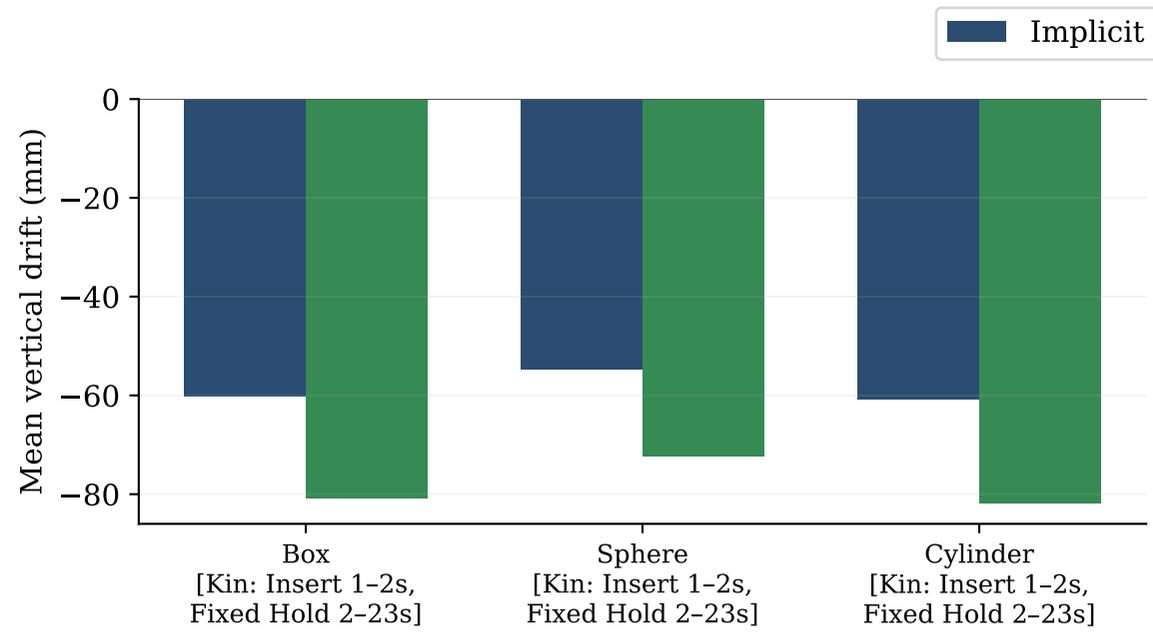
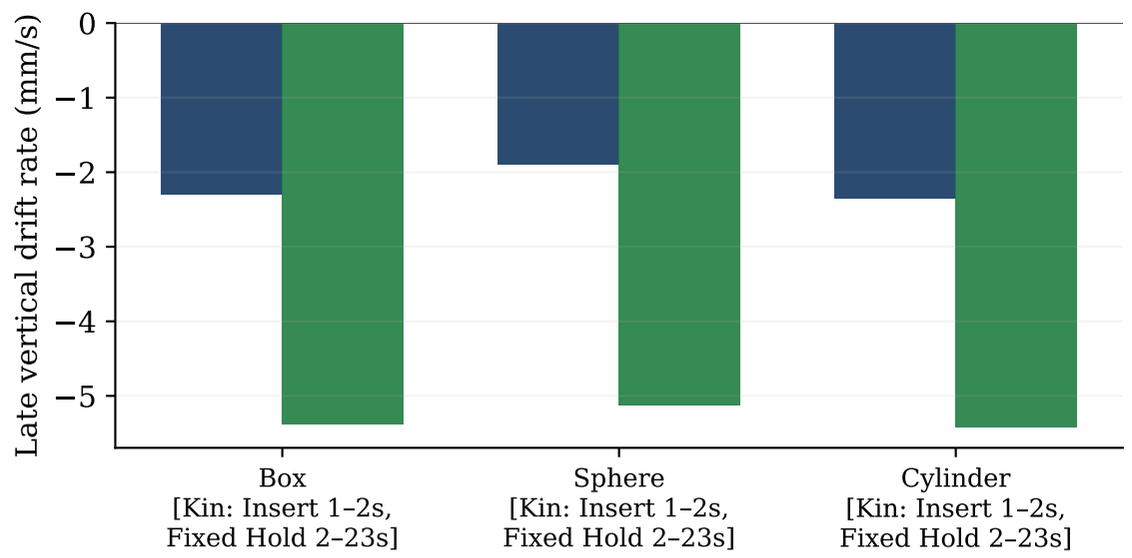
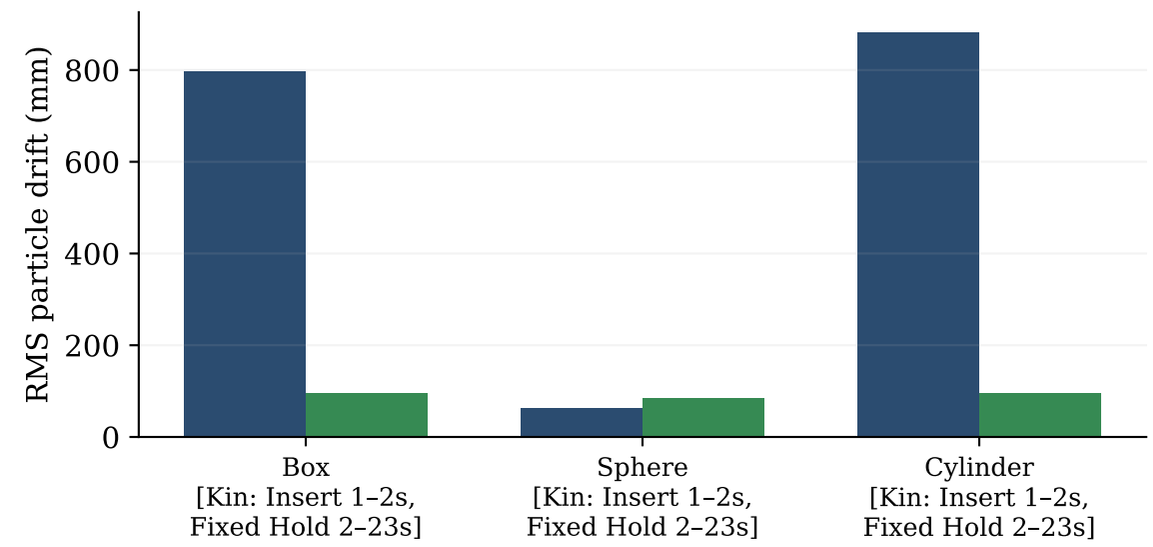
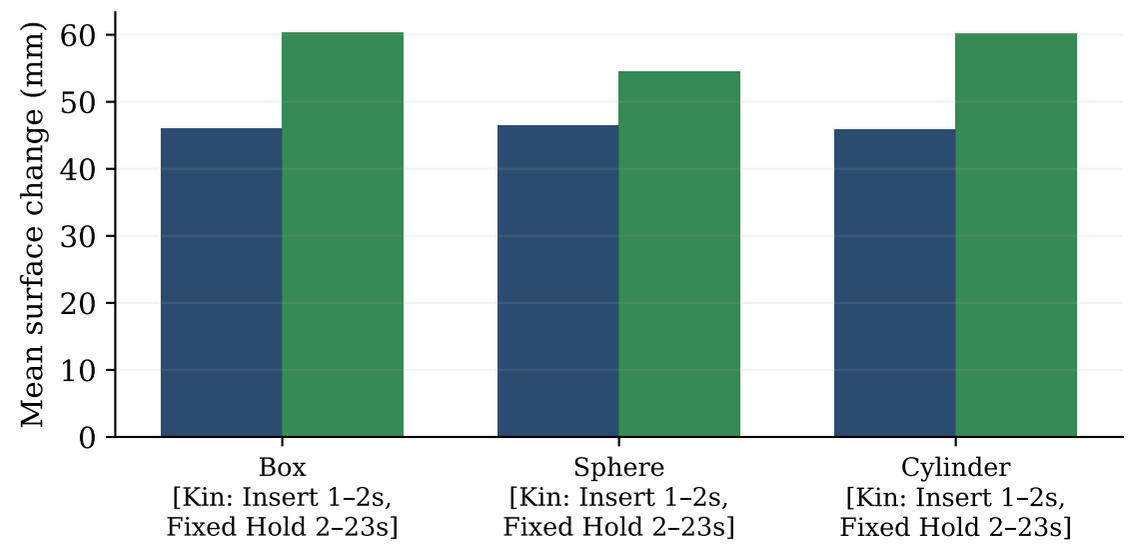
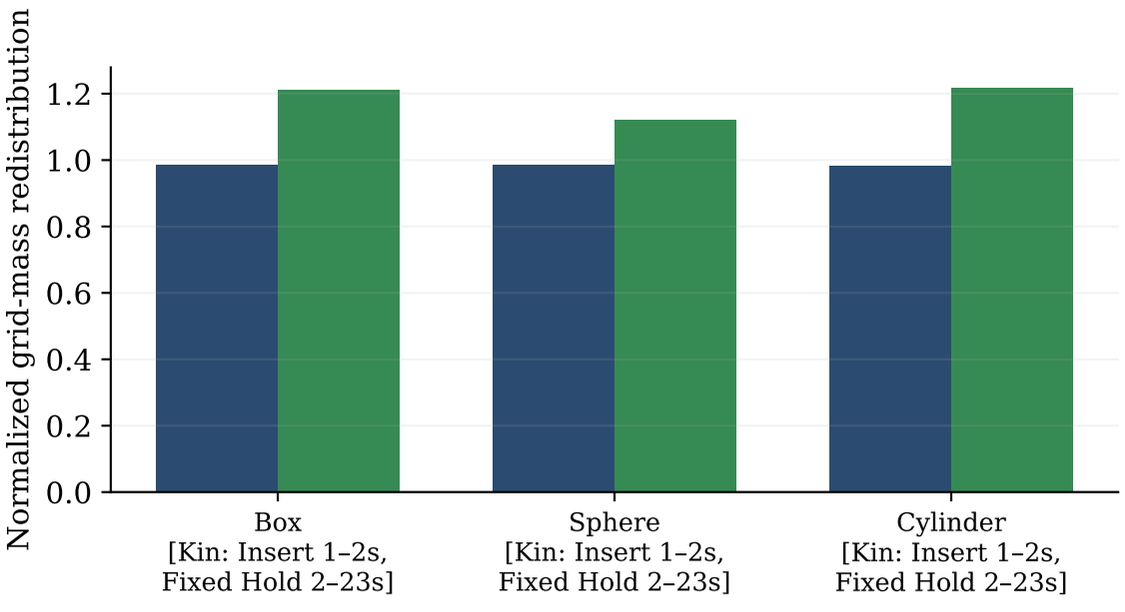
Long-Horizon Simulation Evaluation (23 s): Kinematic Vertical Insertion (1–2 s) + Fixed-Pose Hold (2–23 s)











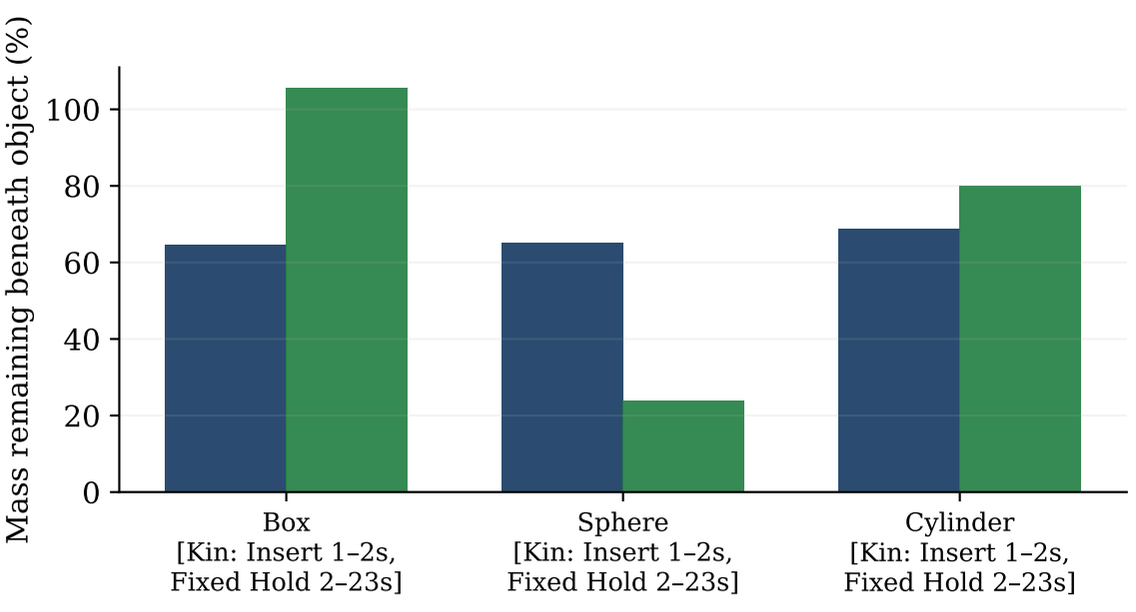


Figure 7: Frozen object-model checkpoint 11,000. Long-horizon 23-second evaluation (Kinematic Motion: Object is inserted vertically from 1–2 s and held at an exactly fixed pose from 2–23 s; measured relative to 3 s). Top row: mean change in world z at 23 s relative to 3 s, and late drift rate (13–23 s). Middle row: RMS particle drift and mean surface change. Bottom row: grid-mass redistribution and mass remaining beneath object.